

1. Rewrite the equation $8x - 12y = 48$ in slope-intercept form.

$$8x - 12y = 48$$

$$8x - 8x - 12y = 48 - 8x$$

$$-12y = -8x + 48$$

$$\frac{-12y}{-12} = \frac{-8x}{-12} + \frac{48}{-12}$$

$$y = \frac{2}{3}x - 4$$

Slope-Intercept Form

$$y = \frac{2}{3}x - 4$$

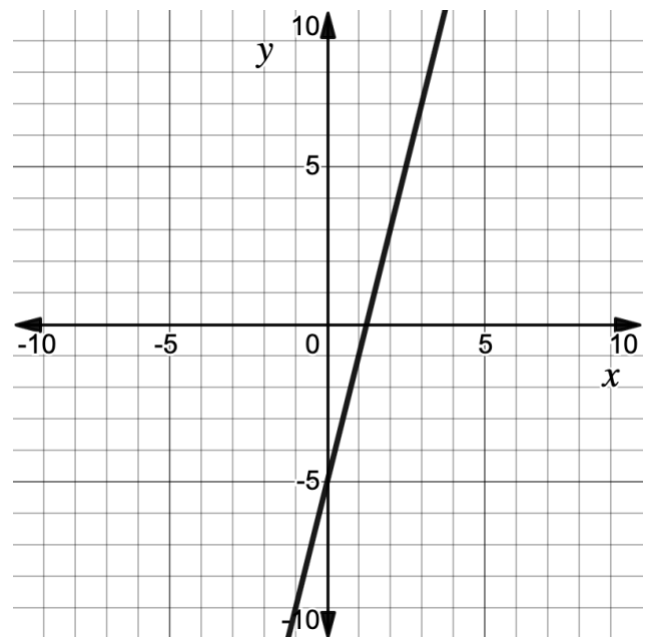
2. The graph of a line is shown on the coordinate grid.

- a. Write the equation of the line in point-slope form.

$$m = \frac{-5 - 3}{0 - 2} = \frac{-8}{-2} = 4$$

Point-Slope Form

$$y - 3 = 4(x - 2)$$



- b. Write the equation of the line using standard form.

$$y - 3 = 4(x - 2)$$

$$y - 3 = 4x - 8$$

$$y - 3 + 3 - 4x = 4x - 4x - 8 + 3$$

$$y - 4x = -5$$

$$-1(y - 4x) = -1(-5)$$

Standard Form

$$4x - y = 5$$

3. A table of values is shown.

x	-8	-3	2	5
y	8	$4\frac{7}{8}$	$1\frac{3}{4}$	$-\frac{1}{8}$

Write the equation of the line in point-slope form.

$$m = \frac{8 + \frac{1}{8}}{-8 - 5} = \frac{\frac{65}{8}}{-13} = -\frac{65}{104}$$

$$y - 8 = -\frac{65}{104}(x - (-8))$$

Point-Slope Form	$y - 8 = -\frac{65}{104}(x + 8)$
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4. A line passes through the points $(-5, -11)$ and $(3, -11)$. Write the equation of the line using any form.

$$m = \frac{-11 - (-11)}{3 - (-5)} = \frac{0}{8} = 0$$

$$y - (-11) = 0(x - 3)$$

$$y + 11 = 0$$

$$y + 11 - 11 = 0 - 11$$

$$y = -11$$

Equation	$y = -11$
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5. A line passes through the points $(8.2, 15.9)$ and has a slope of 4. Write the equation of the line using any form.

Equation	$y - 15.9 = 4(x - 8.2)$
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6. A line passes through the points (0,6) and (−8,0). Write the equation of the line using any form.

$$m = \frac{0 - 6}{-8 - 0} = \frac{-6}{-8} = \frac{3}{4}$$

$$y - 6 = \frac{3}{4}(x - 0)$$

Equation	$y - 6 = \frac{3}{4}x$
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7. Kés Shaun conducted an experiment using a toy car and ramps of various heights. He measured the distance, in centimeters (cm), the toy car traveled from the end of the ramp. He also recorded the height, in centimeters (cm), of the ramp. He found that the toy car traveled 64 cm when the ramp height was 15 cm. The toy car traveled 112 cm when the ramp height was 26 cm.

Write the equation of the line, in any form, where x represents the ramp height, in cm, and y represents the distance, in cm, the toy car travels.

(15, 64) and (26, 112)

$$m = \frac{112 - 64}{26 - 15} = \frac{48}{11}$$

Equation	$y - 64 = \frac{48}{11}(x - 15)$
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